

$$\Rightarrow y = \frac{1}{5}$$

∴ There exists a unique right multiplication give I. by as

Exercises 2.2

1. Find the inverse of the matrix $\begin{bmatrix} 8 & 6 \\ 5 & 4 \end{bmatrix}$

Solution:

Determinant $d = 8(4) - 6(5) = 2$

this matrix is $\frac{1}{2}$

$$\begin{bmatrix} 4 & -6 \\ -5 & 8 \end{bmatrix} = \begin{bmatrix} 2 & -3 \\ -5/2 & 4 \end{bmatrix} \cdot \square$$

By Theorem 4, the inverse of

5. Use the inverse found in Exercise 1 to solve the system

$$8x_1 + 6x_2 = 2$$

$$5x_1 + 4x_2 = -1$$

Solution:

$$\begin{bmatrix} 8x_1 + 6x_2 \\ 5x_1 + 4x_2 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 8x_1 \\ 5x_1 \end{bmatrix} + \begin{bmatrix} 6x_2 \\ 4x_2 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$\Rightarrow x_1 \begin{bmatrix} 8 \\ 5 \end{bmatrix} + x_2 \begin{bmatrix} 6 \\ 4 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} 8x_1 \\ 5x_1 \end{bmatrix} + \begin{bmatrix} 6x_2 \\ 4x_2 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

(Defn. of vector addition)

(Defn. of scalar multiplication)